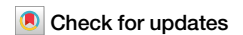


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Rapid urban land uplift enhanced by crustal faults during groundwater recovery



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Urban land subsidence driven by groundwater extraction is widespread, but the processes governing how groundwater recovery translates into spatially variable surface uplift remain poorly understood. We use satellite geodesy and multi-decadal groundwater observations to measure vertical land motion in Greater Osaka, Japan, where aquifers are intersected by a dense network of crustal faults. We observe widespread surface uplift at rates up to 12 mm/year, organized into distinct domains separated by sharp boundaries. Uplift rates vary abruptly across domain boundaries, which coincide with mapped faults. Where no faults are mapped, abrupt spatial changes in uplift rate suggest the presence of previously unrecognized buried faults. Uplift rates are positively correlated with groundwater-level rise in wells screened at depths up to 600 m, while deformation modelling indicates dominant uplift sources are shallower than 500 m. Our analysis suggests crustal faults act as hydraulic barriers to lateral groundwater flow, localising vertical land motion during aquifer recovery, with implications for groundwater management, infrastructure stability, and land-use planning in urban basins worldwide.

Land subsidence affects many of the world's major metropolitan regions and represents a growing hazard to urban infrastructure, ecosystems, and human safety^{1–5}. The impacts are particularly severe in coastal cities, where subsidence amplifies relative sea-level rise and increases exposure to flooding and long-term damage^{5–8}. At the global scale, groundwater abstraction is widely recognized as the dominant driver of urban subsidence^{5,9,10}. Consequently, improved groundwater management is often promoted as a primary strategy for mitigating or reversing subsidence^{1,11}. Despite these efforts, groundwater levels in most major aquifers continue to decline^{9,12}, and sustained recovery following management interventions remains uncommon¹³.

Understanding how urban landscapes respond during sustained groundwater recovery is therefore critical. The mechanical relationship between groundwater-induced pore-pressure changes and land-surface deformation is well established, with groundwater extraction and recovery producing subsidence and uplift responses through changes in aquifer-system stress and compaction^{14–23}. Similar mechanical responses have also been documented across a range of engineered subsurface systems, including hydrocarbon reservoirs and fluid injection operations, where ground-surface deformation provides insight into subsurface pressure changes^{24–29}. Where sustained, multi-annual groundwater recharge has occurred^{15,21,30–34}, surface uplift is often spatially heterogeneous and typically

recovers only a fraction of prior subsidence due to irreversible aquifer compaction. In sedimentary basins that host major cities, groundwater flow and pressure evolution are influenced not only by pumping history but also by subsurface heterogeneity, including crustal faults that can partition groundwater systems and modify hydraulic connectivity^{35–42}. However, the extent to which faults control the magnitude and spatial variability of surface uplift during long-term groundwater recovery remains poorly understood.

Greater Osaka, Japan, provides a rare natural laboratory to investigate the surface deformation response to long-term groundwater recovery within a major urban aquifer system that is intersected by a network of active crustal faults. The region, which includes the Osaka Plain and the Kyoto and Nara basins, hosts a population of approximately 15 million that overlies an extensive groundwater system exceeding 3000 km². Intensive groundwater abstraction between the 1920s and 1960s lowered groundwater levels by up to 30 m and caused cumulative land subsidence exceeding 2.0 m across much of the basin, prompting the introduction of strict groundwater regulations in the early 1960s^{43,44}. Since then, groundwater levels have risen over multi-decadal timescales, while ground subsidence largely stabilized by the mid-1970s. Previous space-geodetic studies have reported localized surface uplift within Greater Osaka^{45,46}, but the governing mechanisms and potential links to groundwater have remained unclear.

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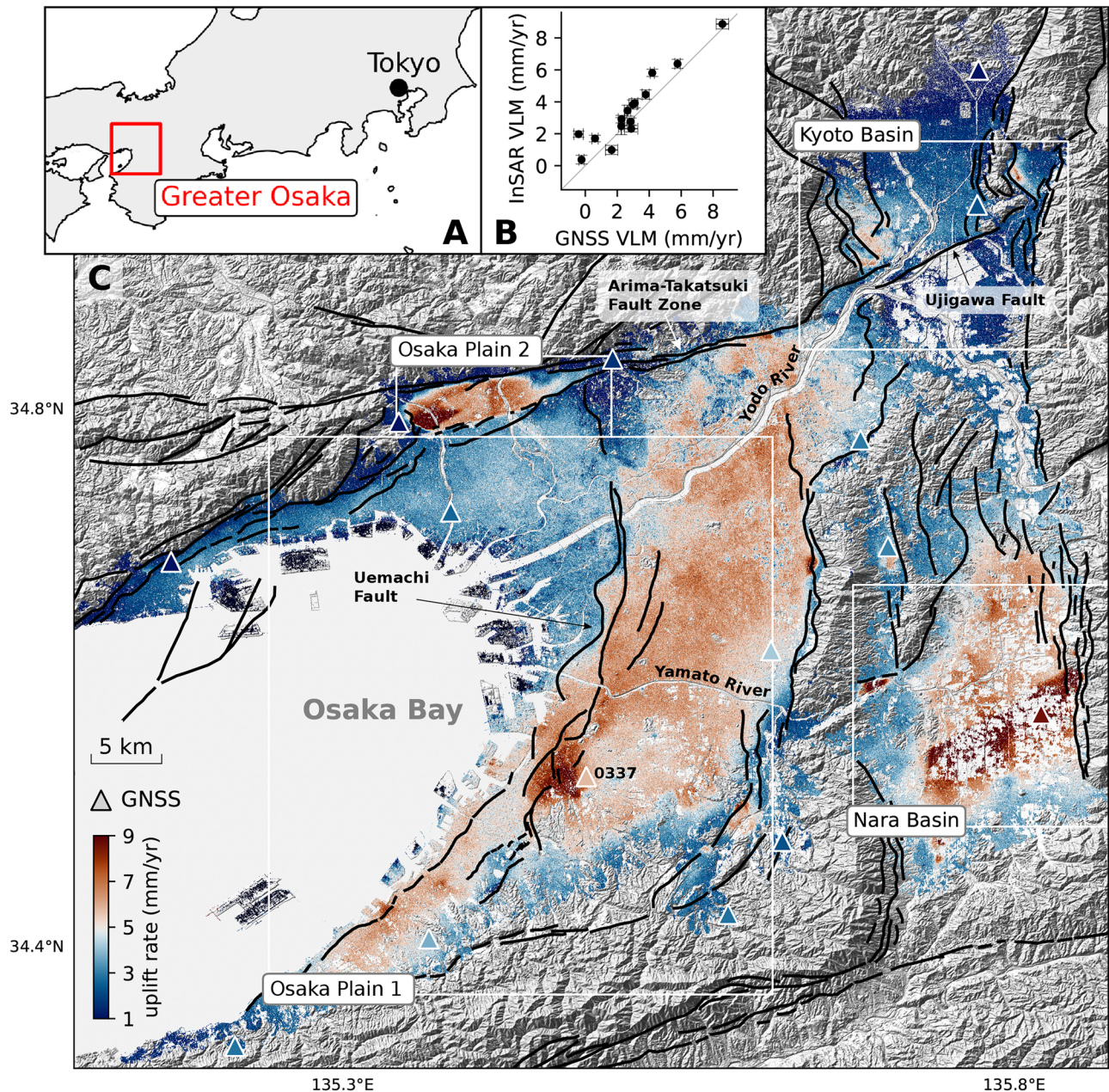


Fig. 1 | Vertical land motion (VLM) in Greater Osaka. **A** Map of Japan showing the study area (red box) shown in (C). **B** The correlation between VLM measured by InSAR and GNSS at $n=15$ stations shown in (C). Measured velocities are calculated from GNSS and InSAR timeseries data, while the error bars represent 1σ uncertainties. **C** Map of InSAR-derived VLM in Greater Osaka. Vertical velocities are

relative to the International Terrestrial Reference Frame 2014. GNSS stations are coloured at the same scale as InSAR-derived VLM; GNSS station 0337 is shown. Black shading near the coast denotes subsidence. Black lines represent fault traces based on the AIST active fault database⁶⁸. The white boxes are the areas shown in Figs. 2 and 3.

Here we combine multi-decadal groundwater level records with high-resolution satellite geodetic observations to document rapid and spatially heterogeneous surface uplift during sustained aquifer recovery. We show that crustal faults exert a first-order control on groundwater accumulation and vertical land motion by impeding subsurface flow and localizing aquifer expansion. These observations capture the state of an urban aquifer system more than 60 years after groundwater regulation was implemented and provide insight into how cities worldwide may evolve as they transition from subsidence to recovery.

Results

Surface uplift in Greater Osaka

High-resolution geodetic observations reveal widespread yet sharply segmented surface uplift across Greater Osaka (Fig. 1). Using

interferometric synthetic aperture radar (InSAR) time-series analysis of 293 Sentinel-1 images acquired between December 2016 and December 2021, together with co-located GNSS observations, we measure the vertical and east-west deformation. InSAR captures relative deformation patterns with high spatial detail, while GNSS provides absolute positioning within a stable reference frame, allowing us to measure surface displacements at 10 m spatial resolution (Methods). Note that we explicitly do not apply multilooking in our processing of the SAR data in order to preserve short-wavelength spatial variations in deformation. The resulting map shows widespread uplift across the urban region, with a mean vertical rate of $4.0 \pm 2.5 \text{ mm yr}^{-1}$, consistent with uplift due to the earthquake cycle within the Nankai subduction zone⁴⁷. Uplift is organized into broad domains exceeding 100 km^2 with relatively uniform rates, but these domains are separated by sharp

boundaries across which uplift rates change abruptly over short distances of less than 100 metres.

Within the Osaka Plain, a pronounced north to northeast trending boundary, less than 100 m wide, separates two distinct uplift domains and coincides with the mapped trace of the Uemachi Fault, a major east-dipping reverse fault (Fig. 1). Average uplift rates on the eastern, hanging-wall side of the fault reach 6 to 7 mm yr⁻¹, compared to 3 to 4 mm yr⁻¹ on the western side. Bedrock uplift rates bordering the western margin of the Osaka Plain are only 0–1 mm yr⁻¹, implying that a substantial component of the observed uplift arises from basin-scale processes rather than regional tectonic uplift. Near the western margin of the Osaka Plain, a localized wedge-shaped zone smaller than 25 km² displays rapid uplift of 7 to 10 mm yr⁻¹ and is bounded by active strands of the Arima-Takatsuki Fault Zone (Fig. 1). In the Kyoto Basin, a similarly sharp gradient in uplift rate aligns with the Ujigawa Fault (Fig. 1). Across these major structures, the average difference in uplift rate is approximately 6.5 mm yr⁻¹. In contrast, the Nara Basin exhibits a prominent zone of uplift reaching up to 12 mm yr⁻¹, with lower rates in the northern part of the basin. This transition occurs across a sharp (<100 m wide), 12-km-long northeast-trending linear boundary that is not associated with any previously mapped fault (Fig. 1).

Horizontal surface motion provides additional insight into the deformation pattern. East-west velocities display a broad northwest to southeast gradient across the region, consistent with long-wavelength right-lateral plate boundary strain (Fig. S1B). Superimposed on this regional signal, many of the north-trending margins of uplifting domains are marked by narrow corridors, 0.1 to 1 km wide, of enhanced east-west deformation. These corridors show westward motion on the western side and eastward motion on the eastern side, forming localized shear zones along uplift boundaries (Fig. S1B). Along the Uemachi Fault, this pattern manifests as a narrow strip of ground moving westward by up to 2 mm yr⁻¹ relative to the surrounding surface.

Time series of line-of-sight deformation from both ascending and descending satellite orbits indicate steady uplift in the Osaka Plain and Kyoto Basin throughout the observation period, superimposed on seasonal variability (Fig. S1F). Continuous GNSS measurements extend this record within the Osaka Plain back to 1996, demonstrating that uplift has persisted for at least the past 28 years (Fig. S2). Based on this, we infer that the spatial pattern of deformation imaged by our joint InSAR-GNSS data reflects long-term uplift of the Osaka Basin. In contrast, time series from the Nara Basin show that uplift initiated more recently, beginning in mid-2017 and continuing at an approximately constant rate thereafter (Fig. S1F).

Multi-decadal groundwater variations

Groundwater in Greater Osaka is hosted within a thick, laterally extensive, confined aquifer system that records multi-decadal changes in subsurface pore pressure. Aquifers in the Osaka Plain and the Kyoto Basin are developed within Plio-Pleistocene sediments of the Osaka Group^{44,48}. These aquifers consist primarily of coarse gravel and sand units hydraulically confined by interbedded low-permeability marine clays (aquitards), forming a multilayered system with a total thickness of approximately 800 m. To characterize long-term groundwater dynamics, we analyse groundwater head changes from 1985 to 2025 using data from 44 monitoring wells screened within Osaka Group gravels at depths between 10 and 561 m (Methods). Variations in groundwater head reflect changes in pore pressure within the confined aquifer system^{39,49}. Several wells are screened across multiple aquifers, allowing groundwater head changes to be resolved at different depths. While some wells record different rates of groundwater-head change at different screening depths, we observe no systematic relationship between groundwater head changes and screening depth across the monitoring network as a whole (Fig. S3). Over the observation period, wells record substantial groundwater level changes, with decadal-averaged rates reaching up to 1.3 m yr⁻¹ and cumulative amplitudes of as much as 31 m.

Groundwater level changes exhibit a systematic spatial organization relative to major crustal faults across Greater Osaka (Fig. 2). In particular, groundwater levels increase more rapidly on the upstream sides of faults

than on their downstream sides, where downstream refers to the lower-elevation side of the basin toward which regional groundwater flow is directed. This pattern is observed repeatedly across independent fault-bounded regions, suggesting that faults exert a strong control on subsurface hydraulic connectivity and pressure evolution.

Within the Osaka Plain, groundwater recovery is widespread but strongly asymmetric across the Uemachi Fault. All 31 analysed wells show net groundwater level increases over the 40-year observation period, indicating basin-wide recovery of pore pressure within the confined aquifers (Fig. 2A). However, wells located on the eastern, upstream side of the Uemachi Fault record an average groundwater level rise of 0.3 m yr⁻¹, compared to 0.1 m yr⁻¹ on the western side. This asymmetry suggests that the Uemachi Fault impedes lateral groundwater flow, promoting pore pressure accumulation on its upstream side. Consistent with this interpretation, a marked decline in groundwater head between 1991 and 1998 is observed at 8 of 9 wells west of the fault, associated with large-scale underground construction⁵⁰, while no comparable decline is detected at wells east of the fault (Fig. 2A).

A similar fault-controlled contrast in groundwater behaviour is observed in the Kyoto Basin across the Ujigawa Fault. Groundwater levels from 12 wells spanning 2000 to 2023 reveal distinct trends on either side of the fault (Fig. 2B). Wells located north of the fault, on the upstream side, show an average groundwater level increase of 0.54 m yr⁻¹. South of the fault, groundwater levels increase only weakly, at an average rate of 0.01 m yr⁻¹, and several wells record declining groundwater levels, indicating net decreases in pore pressure. The most pronounced recovery occurs at well NKY on the northern side of the Ujigawa Fault, where groundwater level increased by 23.4 m between 2000 and 2010, corresponding to rates of approximately 2.3 m yr⁻¹.

Temporal changes in groundwater trends within the Arima-Takatsuki Fault Zone reveal dynamic fault-scale control on pore pressure evolution. At well IKD, located within a wedge-shaped zone of rapid surface uplift, groundwater levels declined between 1998 and 2012 at an average rate of 0.16 m yr⁻¹ before reversing to a rising trend of 0.23 m yr⁻¹ that persisted until 2019 (Fig. 3A, B). In contrast, a nearby well (well TKZ) located outside both the fault zone and the region of rapid uplift shows no such reversal and instead records a steady groundwater level increase of 0.07 m yr⁻¹ between 1997 and 2025 (Fig. 3B). During the period covered by the InSAR observations from December 2016 to December 2021, groundwater levels at this well show no clear trend.

Spatial variations in groundwater level rise closely mirror the pattern of surface uplift observed across Greater Osaka (Fig. 2C). Average groundwater level increases on the downstream sides of major faults are 0.06 m yr⁻¹, corresponding to average uplift rates of 2.2 mm yr⁻¹. In contrast, upstream sides of faults experience faster groundwater level increases of approximately 0.30 m yr⁻¹ and higher uplift rates of 6.0 mm yr⁻¹. Although these observations alone do not establish a causal relationship, the systematic correspondence between subsurface pore-pressure evolution and surface deformation supports a mechanical linkage between groundwater recovery and vertical land motion, potentially mediated by major crustal faults within the basin.

Modelling of aquifer expansion

To investigate the processes responsible for the observed surface uplift, we compare a set of simple elastic deformation models with the geodetic observations. Potential sources of multi-annual uplift include aseismic creep on shallow-dipping reverse faults and expansion of subsurface aquifers driven by increases in pore pressure. We therefore evaluate three classes of models, an elastic slip model, an elastic opening model, and an elastic distributed-source model, and assess how well their predicted deformation fields match the observed vertical and horizontal land motion (Methods).

We first examine whether shallow fault creep could explain the deformation observed across major faults. Although aseismic creep has not been reported on onshore faults in the Greater Osaka region, we apply an Okada-type slip model to the Uemachi Fault as a representative case

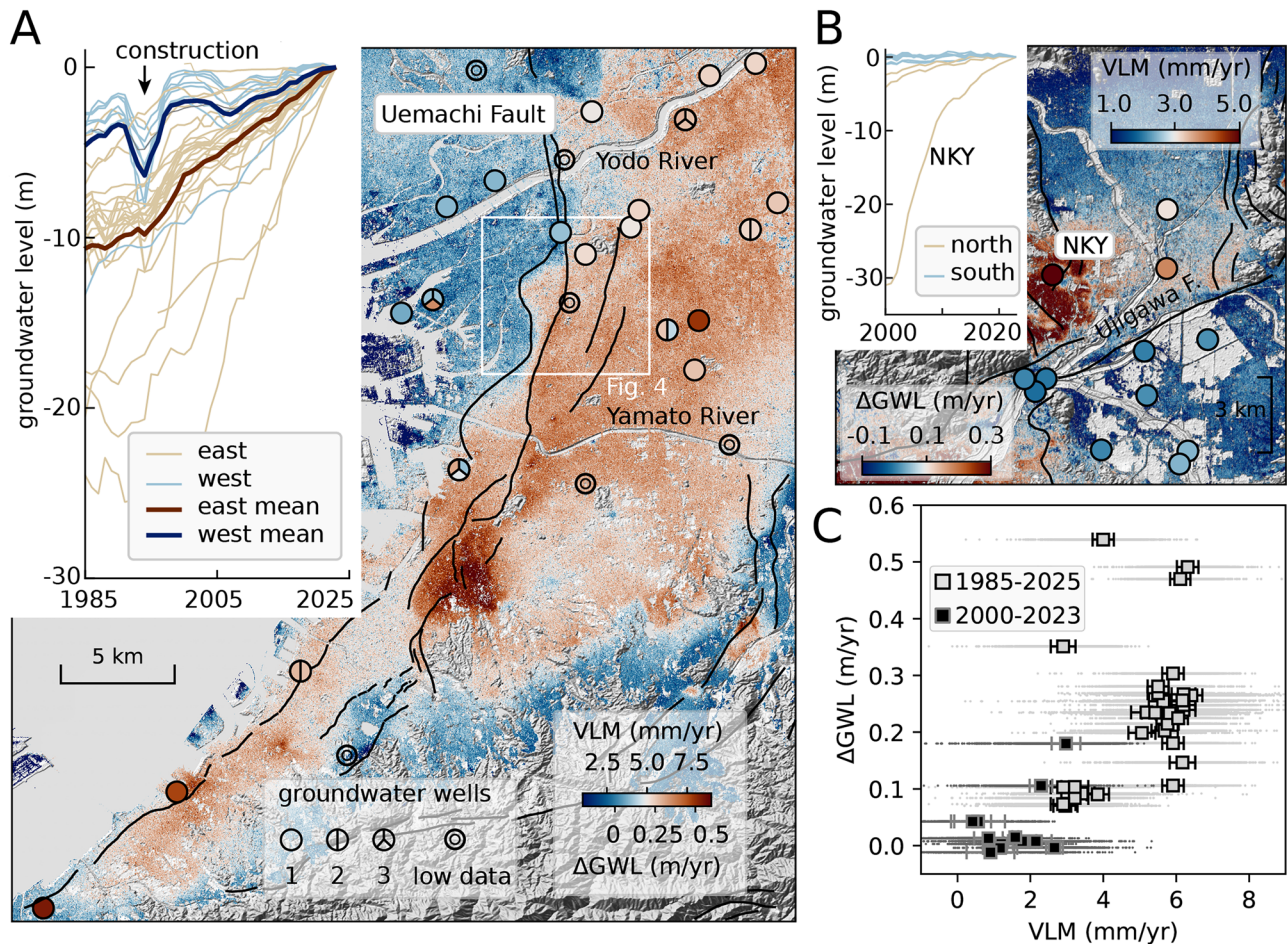


Fig. 2 | Relationship between VLM and groundwater level (GWL) change across the Uemachi fault. **A** Groundwater changes and surface uplift within Osaka Plain. Wells (circles) are coloured according to the rate of GWL rise from 1985 to 2025. Wells lacking data in 1985 are shown with double circles. Wells with multiple screening depths are plotted as pie charts with corresponding screening depth shown in Fig. S3. The background shows InSAR-derived VLM. Black lines denote faults from the AIST active fault database⁶⁸. Time-series data show relative groundwater head changes for all wells. The colour of the time-series data shows whether the well is located east (red) or west (blue) of the Uemachi Fault. Wells located within the

distributed fault zone of the Uemachi fault are assigned to the east or west of the Uemachi fault based on our InSAR-derived uplift pattern. Thick dark lines represent mean groundwater changes. **B** Same as for **A**, but for the Kyoto Basin wells surrounding the Ujiyawa Fault. **C** Comparison of rate of GWL change and vertical land motion. Solid squares show the mean and 1σ of all VLM measurements within 500 m of each well. $n = 93809$ individual VLM measurements were used across all wells. For wells with multiple screening depths, rates are shown for the shallowest screening depth.

(Fig. S4). The model reproduces the sense of vertical uplift across the fault, but it also predicts pronounced east–west convergence normal to the fault that is not observed in the InSAR-derived deformation field (Fig. S4). Because such convergence is an inherent feature of shallow-dipping reverse fault slip, regardless of fault depth, geometry, or slip rate, we conclude that fault creep cannot account for the observed uplift and do not consider this mechanism further.

We next evaluate aquifer expansion as a potential source of surface uplift by modelling deformation induced by volumetric strain at depth. In this approach, surface deformation is calculated as the elastic response of a half-space to either a planar opening source, following Okada-type tensile dislocation theory⁵¹, or to a three-dimensional distributed source that represents volumetric strain within a deforming reservoir⁵². For the distributed-source model, deformation is controlled by the imposed volumetric strain, which we assume to be laterally uniform with equal horizontal components ($e_{11} = e_{22}$). The ratio of vertical to horizontal strain, e_{33}/e_{11} , is treated as a free parameter α , allowing for anisotropic deformation associated with structural confinement. For both aquifer-based models, source magnitudes are estimated using non-negative least squares. We evaluate model sensitivity through grid searches over source depth at 100 m intervals between 0 and 1000 m depth and, for the distributed-source model, we

explore different values of α . Because our models are spatially uniform in their mechanical properties, they do not account for spatial heterogeneity within the aquifer system.

Models based on aquifer expansion successfully reproduce the first-order characteristics of the observed land motion across multiple fault-bounded regions. Best-fitting examples of the elastic distributed-source model for the Uemachi Fault and the Arima-Takatsuki Fault Zone are shown in Fig. 4, while corresponding results for the Nara and Oji regions are presented in Fig. S5. In all four areas, the models capture both the amplitude and spatial pattern of vertical uplift as well as the associated horizontal deformation. Our sensitivity analysis demonstrates that models with source depths shallower than approximately 500 m consistently provide better fits to the observed deformation than deeper sources (Figs. S5 and S7). While deeper aquifers may also contribute to the observed uplift, the grid-search results indicate that they are unlikely to be the dominant source of deformation. In particular, our sensitivity analysis shows that edges of the uplifting regions, where uplift rates vary spatially over short distances, require source depths to be shallow, while broad, spatially uniform uplift provides little constraint on source depth (Figs. S5 and S7). Our modelling approach with spatially uniform properties does not capture the variability observed within fault-bounded aquifer domains, which is likely to be related

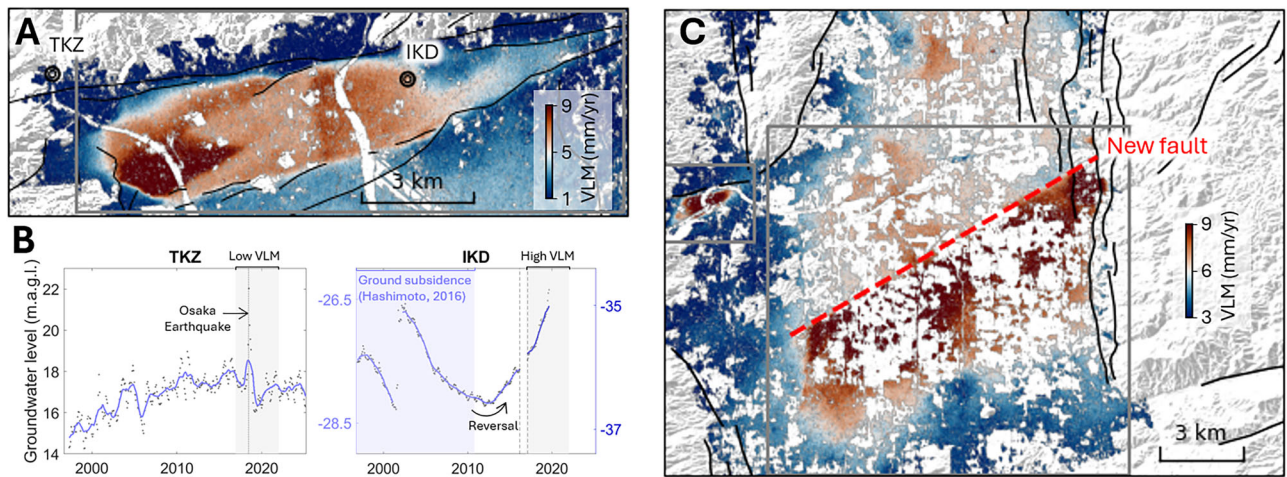


Fig. 3 | VLM and GWL changes at the Arima-Takatsuki Fault zone, and newly proposed fault in the Nara basin. **A** Map of InSAR-derived VLM around the Arima-Takatsuki fault zone (the area labelled as Osaka Plain 2 in Fig. 1C). The double circles show the location of wells. Black lines show fault traces based on the AIST active fault database⁶⁸. The gray box is the area shown in Fig. 4. **B** Groundwater time series in metres above ground level (aquifer is pressurised) at the wells shown in panel A. Gray dots indicate monthly means, and solid lines show 12-month moving averages. Vertical dashed black lines mark the dates of the exchange of water gauges.

The timing of the 2018 M6.1 Osaka earthquake, which caused a transient groundwater-level increase, is also indicated. Well IKD records a transition from GWL decline to GWL rise in 2012, which is consistent with the change from subsidence to uplift measured by InSAR. **C** Map of InSAR-derived VLM in the Nara Basin (the area labelled as Nara Basin in Fig. 1C). Black lines show fault lines based on the AIST active fault database⁶⁸, and the red dashed line indicates the newly proposed fault. The gray box indicates the region enlarged in Fig. S5.

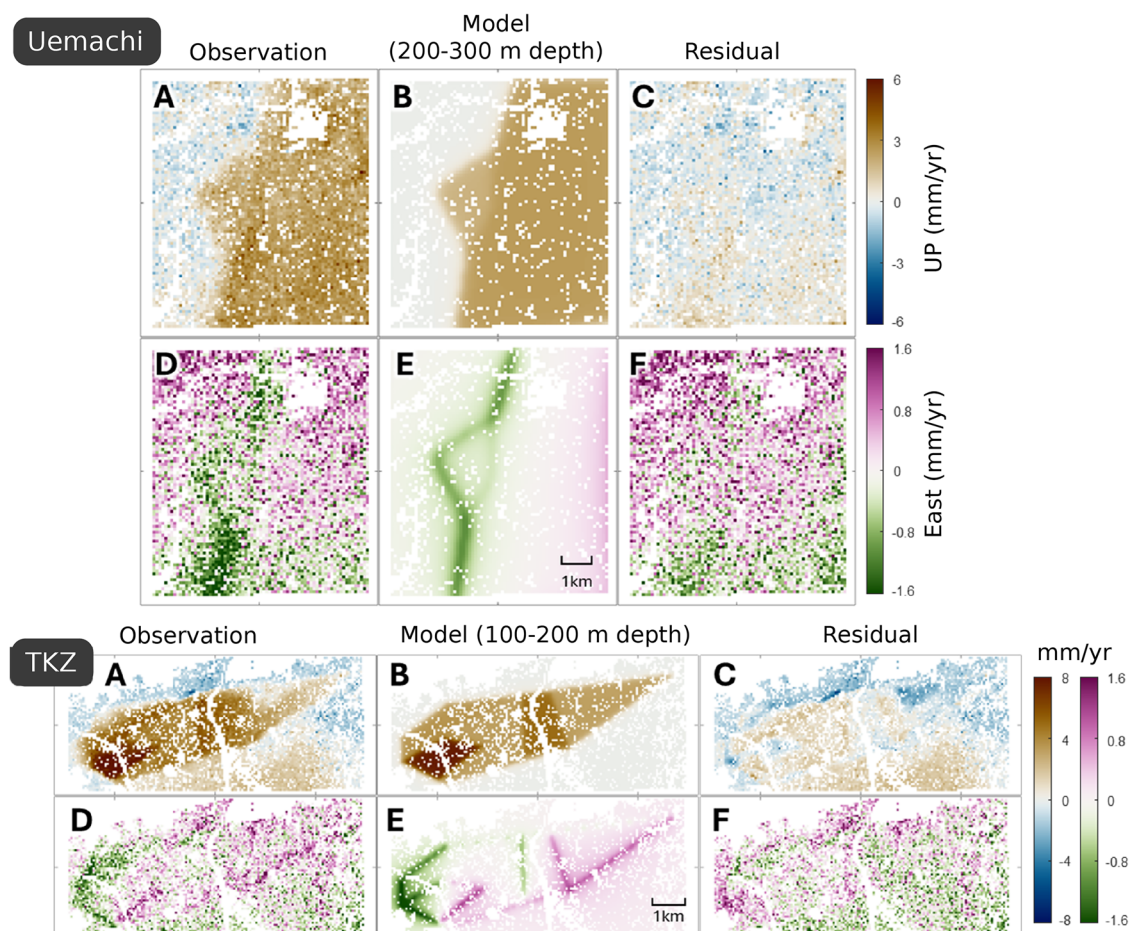


Fig. 4 | Model results for the Uemachi Fault and Arima-Takatsuki fault zone (TKZ). Comparison between observed and modelled deformation fields. **A, D** InSAR-derived vertical land motion (VLM) and east-west deformation observations. **B, E** Elastic distributed-source model results obtained from least-squares inversion. The depths of the top and bottom surfaces of the distributed-

source models are shown. **C, F** Residuals between observations and model predictions (observations minus model). The spatial distributions of sources representing expanding aquifers are shown in Fig. S7 for the Uemachi Fault and Fig. S5 for the TKZ.

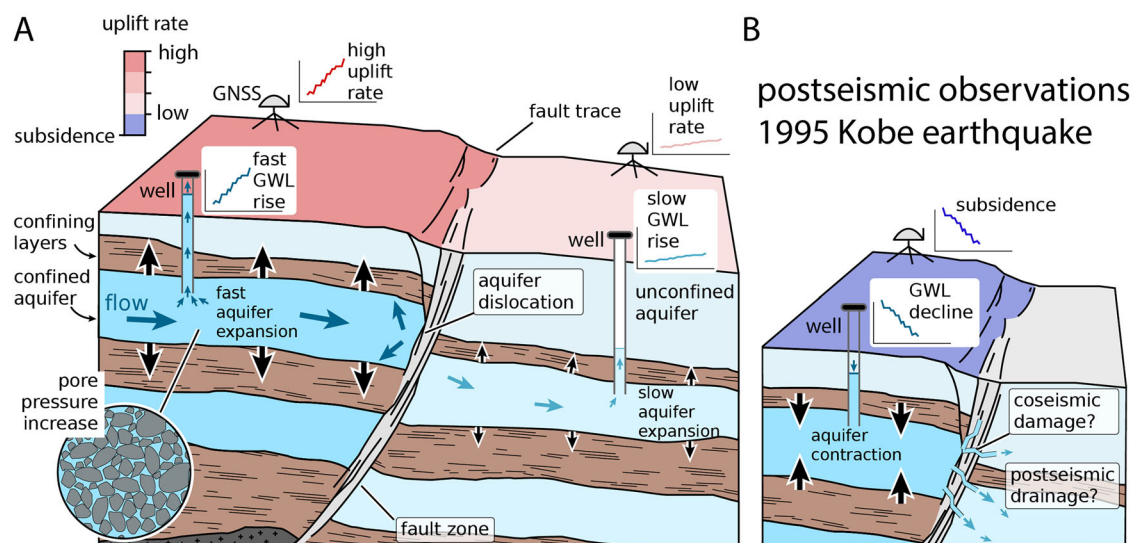


Fig. 5 | Conceptual model of deformation. Schematic illustration showing relationship between groundwater level, aquifer deformation and surface uplift rate. **A** Lateral gravity-driven flow of groundwater within a confined aquifer becomes impounded against the fault zone, resulting in larger increases in groundwater level on the upstream side of the aquifer, and leading to reduced groundwater level rise on the other side of the fault. This leads to differential elastic expansion of the aquifer across the fault, resulting in observed differences in uplift rate across the fault, with a

sharp gradient at the fault trace. **B** Postseismic deformation within the Arima-Takatsuki Fault Zone following the 1995 Kobe earthquake. Rapid subsidence (-5 to -10 mm yr^{-1}) and groundwater level decline (-0.2 m yr^{-1}) were observed between 1998–2010, suggesting drainage and contraction within the confined aquifer. Since 2013 groundwater level is rising (0.2 m yr^{-1}) and rapid uplift (up to 10 mm yr^{-1}) is observed, suggesting groundwater trapping and aquifer expansion have recommenced.

to lateral changes in lithology or permeability within the aquifer. Although our simplified approach does not explicitly model deformation related to groundwater flow, our results indicate that elastic expansion of fault-bounded aquifers provides a physically consistent explanation for the observed land motion.

Discussion

The surface uplift observed across Greater Osaka can be explained by fault-controlled groundwater trapping and the resulting elastic expansion of confined aquifers. Gravity-driven groundwater flow toward Osaka Bay becomes impounded where crustal faults disrupt hydraulic connectivity, leading to higher rates of groundwater-level rise on the upstream sides of faults (Fig. 5A). Accumulation of groundwater increases pore pressure within confined aquifers, which respond by expanding elastically and uplifting the ground surface above. This mechanism accounts for the sharp, fault-aligned gradients in uplift rate observed across the region and explains why uplift is spatially organized into discrete fault-bounded domains.

A simple scaling argument shows that the observed magnitude of uplift is consistent with elastic expansion driven by groundwater recovery. Assuming hydrostatic conditions, an increase in groundwater head of order 0.1 to 0.3 m yr^{-1} corresponds to a pore-pressure increase of approximately 1 to 3 kPa yr^{-1} ¹⁵³. For a confined aquifer several hundred metres thick responding elastically under undrained conditions, and adopting representative undrained bulk moduli for unconsolidated to weakly lithified sedimentary deposits ($K_u \sim 1$ – 10 GPa), this pressure increase implies volumetric strain rates on the order of 10^{-6} to 10^{-5} yr^{-1} ¹⁵³. When integrated over the thickness (100 – 600 m) of the deforming aquifer, these strain rates yield surface uplift of several millimetres per year, consistent with the InSAR observations. This order-of-magnitude agreement supports a causal link between observed groundwater-level rise and surface uplift. Our models of aquifer expansion assume homogeneous mechanical properties and therefore do not explicitly represent the layered structure of the aquifer system. Short-wavelength vertical motions observed within broad uplift domains likely reflect heterogeneity associated with interbedded aquifer and aquitard units or other subsurface complexity. Such mechanical contrasts

may locally modify deformation patterns and introduce uncertainty into our model of pressure-induced uplift.

Fault control on aquifer-related surface deformation has been documented in other sedimentary basins, but the mechanism operating in Greater Osaka is distinct from most previous examples. In many fault-bounded basins, faults primarily influence deformation by shaping aquifer architecture during syn-tectonic sedimentation, such that present-day fault traces coincide with high-strain basin margins^{30,37,39,42,54}. In contrast, the deformation patterns observed in Greater Osaka indicate that faults actively regulate lateral groundwater transmission during aquifer recovery. A comparable mechanism has been inferred for the Silver Creek Fault in the Santa Clara Valley, where faults act as hydraulic barriers that compartmentalize groundwater flow³⁹. In Greater Osaka, differential groundwater accumulation across faults may arise from low permeability of rocks within the fault core (e.g., fault gouge), or from abrupt lithofacies changes across the fault plane due to fault slip, or both. Based on our observations, we cannot distinguish between these two scenarios. However, borehole and seismic reflection data indicate up to 300 m of offset across the Uemachi Fault, suggesting that aquifer dislocation contributes to reduced hydraulic connectivity (Fig. S6).

Time-dependent groundwater and deformation signals within the Arima-Takatsuki Fault Zone further indicate that fault hydraulic properties can evolve over time. A wedge-shaped fault-bounded region that is currently uplifting at rates of up to 10 mm yr^{-1} previously experienced rapid subsidence following the 1995 Kobe earthquake (Fig. S8). This region lies near the termination of the coseismic rupture and likely underwent strong shaking and damage. The postseismic transition from subsidence to uplift coincides with a reversal from groundwater-level decline to groundwater-level rise that occurred in 2012 (Fig. 3B), and was not accompanied by any changes in seismicity rate (Fig. S9). One plausible explanation is that coseismic and early postseismic increases in fault-zone permeability enabled drainage of trapped groundwater into the surrounding basin, producing subsidence between 1995 and 2010 (Fig. 5B). Subsequent fault-zone healing may have progressively reduced permeability, re-establishing a hydraulic barrier that allowed groundwater to accumulate once again and drive renewed uplift (Fig. 5A). These observations suggest that fault permeability,

and its temporal evolution following earthquakes, can exert a strong control on long-term aquifer deformation.

High-resolution geodetic observations also provide a means of identifying buried faults that are not expressed at the surface. In the Nara Basin, we observe a sharp, 12-km long linear gradient in uplift rate aligned along a northeast trend that does not coincide with any mapped surface fault (Fig. 3C). Abrupt lithological changes within the aquifer associated with sediment deposition could also contribute to localized gradients in uplift. However, the pronounced linearity of this boundary and its orientation at a high angle to the basin drainage are difficult to explain by lithological heterogeneity alone and are consistent with the presence of a previously unrecognized fault. Although this structure lacks a topographic expression, independent borehole data support the presence of a northeast-trending subsurface discontinuity in this location⁵⁵. In basins with rapid sedimentation, where surface signs of faulting are weak or absent, groundwater-induced deformation may therefore provide a sensitive indicator of buried fault structures.

These results have important implications for groundwater management and future surface deformation in urban regions worldwide. Greater Osaka provides a rare example of a megacity observed more than six decades after the implementation of effective groundwater regulation, demonstrating that the cessation of excessive extraction can lead to sustained aquifer reinflation and measurable surface uplift over multi-decadal timescales. Although much of the initial subsidence will not be recovered due to permanent compaction of the aquifer, our results show that sustained recovery is possible and is strongly modulated by crustal fault networks, which act as long-lived hydraulic barriers that regulate pore-pressure evolution and localize vertical land motion. This mechanism explains both the magnitude and sharp spatial gradients of uplift observed across the region and highlights the fundamental role of fault systems in governing the long-term deformation of urban basins. As many cities continue to pursue managed aquifer recovery, forecasts of future land motion must account for the hydraulic influence of faults to anticipate spatially heterogeneous uplift and its implications for infrastructure and land-use planning.

Methods

InSAR processing

We use Sentinel-1 Single Look Complex (SLC) images acquired between December 2016 and December 2021 to measure ground deformation in Greater Osaka. All data were obtained from the Alaska Satellite Facility. The dataset comprises 139 images from ascending track 10 and 154 images from descending track 17. To reduce computational cost while maintaining spatial continuity, the study area was divided into two overlapping regions that were processed independently, with the overlap located near the centre of the region of interest. Sentinel-1 SLCs were coregistered and stacked using the InSAR Scientific Computing Environment (ISCE-2) version 2.6.3⁵⁶. Time-series interferometric processing was performed using the software MIAPLPY version 0.2.1⁵⁷, which applies a non-linear phase-linking approach. To avoid fading signal anomalies, we used a single-reference network and full image resolution (i.e., no multilooking was applied). One interferogram from the ascending track (20190904–20190624) was excluded due to poor quality. For both ascending and descending datasets, the reference pixel was set at a continuously operating GNSS station 0761 located at Yao (34.6211°, 135.6188°).

Displacement time series were obtained through network inversion using MINTPY version 1.6.1⁵⁸, followed by corrections for tropospheric and topographic phase delays. Tropospheric delays were corrected using the ECMWF Reanalysis v5 (ERA5) model, and topographic phase contributions were removed using the Copernicus GLO30 digital elevation model. Time-independent LOS velocities were estimated by fitting a linear trend to the displacement time series at each pixel⁵⁹ (Fig. S1C, S1D). The velocity maps were geocoded and masked to exclude pixels with temporal coherence below 0.5, which primarily correspond to water bodies and densely vegetated areas.

To reference the InSAR velocity fields to a stable terrestrial reference frame, we aligned the LOS velocities with co-located GNSS observations in the International Terrestrial Reference Frame 2014 (ITRF2014). GNSS position time series from December 2016 to December 2021 were analysed to estimate time-independent vertical, northward, and eastward velocities over the same period as the InSAR observations (Fig. S1G). Of the 18 GNSS stations within the study area, 15 were retained for analysis. Stations 0760 and 0365 were excluded due to large data gaps, and station 0333 was excluded due to anomalous vertical fluctuations between 2019 and 2020.

GNSS time series were processed using the software “GAMIT” and “GLOBK” with the IGS absolute phase centre model (igs08.atx), and all positions were expressed in the ITRF2014 reference frame. Velocities and uncertainties were estimated using the software Hector version 2.1⁶⁰, assuming a noise model that combines white noise and power-law noise. Known receiver and antenna changes from the GEONET catalogue were accounted for in the calculation of velocities and uncertainties. Velocity uncertainties incorporate contributions from seasonal signals, time-correlated noise, equipment changes, and the effects of nearby large earthquakes. The three-component GNSS velocities were then projected into the LOS directions of the ascending and descending Sentinel-1 orbits.

For each GNSS station, the mean LOS velocity was computed within a 200-m radius and compared with the corresponding InSAR-derived velocity. A planar ramp was estimated by least-squares fitting to the LOS velocity differences between InSAR and GNSS and was removed from the InSAR maps to produce LOS velocity fields in the ITRF2014 frame. Ascending and descending LOS velocity maps were resampled to a common grid with a spatial resolution of 10 m. For each pixel, the east and vertical component of ground velocity is estimated using a linear least-squares inversion⁶¹:

$$\mathbf{G}_{\text{insar}} \mathbf{m}_{\text{insar}} = \mathbf{d}_{\text{insar}}, \quad (1)$$

where $\mathbf{d}_{\text{insar}}$ is a vector containing the ascending and descending LOS velocities, $\mathbf{G}_{\text{insar}}$ is the LOS geometry matrix, and $\mathbf{m}_{\text{insar}}$ is a vector containing the east and vertical components of ground velocity being solved.

Groundwater data

Groundwater level data were compiled from multiple public monitoring networks. Data for the Osaka Plain were obtained from the National Institute of Advanced Industrial Science and Technology (AIST) and the Research Committee of Groundwater and Geo-environment, while data for the Kyoto Basin were obtained from the Ministry of Land, Infrastructure, Transport and Tourism (MLIT).

The Osaka Plain hosts one of the largest groundwater basins in Japan and is underlain by sedimentary deposits that commonly exceed 1 km in thickness. Groundwater flow in the plain is primarily active within the shallow to intermediate portions of the sedimentary sequence⁴⁴. Alluvial deposits cover much of the region with a typical thickness of approximately 10 m, increasing to as much as 35 m along the coast of Osaka Bay⁶². Previous investigations based on land levelling and groundwater data report that ground uplift in the Osaka Plain occurs within shallow confined layers but is not observed within the unconfined surface alluvium and is negligible below depths of approximately 300 m⁴⁸. Based on these observations, wells screened exclusively within the surface alluvium were excluded from the analysis.

To examine the relationship between groundwater head changes and surface deformation across the Uemachi Fault, we selected wells screened within the Osaka Group gravel layers, which comprise sedimentary units above basement rock excluding the surface-connected alluvium⁵⁰. Wells operated by AIST have screening depths ranging from 187.7 to 561.4 m below the ground surface, ensuring that measurements sample confined aquifers within the Osaka Group.

The Nara Basin is characterized by an average surface elevation of approximately 60 m above sea level. Alluvial deposits cover much of the basin and reach a maximum thickness of about 5 m, underlain by terrace deposits with thicknesses of up to 20 m. The Osaka Group extends

downward to basement rock at depths of approximately 600 m. Groundwater in the basin flows toward the central lowlands and is discharged into the Osaka Plain along the Yamato River⁶³.

In the Kyoto Basin, alluvial deposits form the uppermost stratigraphic unit and overlie diluvial sediments. Alluvium thickness is typically a few metres in the northern basin and increases to approximately 12 m in the southern basin⁶⁴. The Osaka Group is also present above basement rock, with the deepest interface between sedimentary deposits and basement occurring at depths of up to approximately 800 m^{65,66}. To avoid sampling unconfined groundwater within the alluvium, only wells with screening depths of 10 m or greater were included in the analysis for the Kyoto Basin.

For all wells meeting the selection criteria in the Osaka Plain and the Kyoto Basin, groundwater level time series were first converted to annual averages to suppress seasonal variability. Average groundwater level change rates were then calculated for each well. For two wells located within the Arima-Takatsuki Fault Zone, TKZ and IKD, groundwater level records were processed by calculating monthly means followed by application of a 12-month moving average to emphasize long-term trends.

Deformation modelling

To investigate the mechanisms responsible for the observed surface deformation, we compare three classes of elastic models: an elastic slip model, an elastic opening model, and an elastic distributed-source model. Because the observed deformation is predominantly time-independent over the analysis period and the modelled displacements scale linearly with slip, opening, or strain, we directly compare modelled deformation fields with observed annual land motion rates. Both the elastic slip and elastic opening models are implemented using Okada's analytical solutions for elastic dislocations in a half-space⁵¹, with surface displacements calculated using MATLAB implementations of these formulations⁶⁷.

Elastic slip model. The elastic slip model represents fault motion as a dislocation on a planar surface. We apply this model to the Uemachi Fault, which is continuous over a length of approximately 36 km⁶⁸. To approximate along-strike continuity, we model the fault as an infinitely long structure with a down-dip length of 100 km and a north-south strike consistent with the mapped fault trace (Fig. S4). The upper edge of the fault is placed at a depth of 0.5 km, the horizontal projection of the fault plane is set to 5 km, and the slip magnitude is fixed at 2.5 mm yr⁻¹, which reproduces the sharp vertical deformation gradient observed at the fault trace (Fig. S4). We compute vertical and horizontal displacement profiles normal to the fault for a range of dip angles of 35°, 50°, 65°, and 80°, and compare these profiles with the observed deformation across the fault (Fig. S4).

Elastic opening model. The elastic opening model represents deformation as tensile opening of a horizontal plane, which we interpret as elastic expansion of an aquifer. Although this formulation is commonly applied to sill intrusion, it provides a convenient approximation for volumetric expansion at depth. Model parameters are estimated by solving a non-negative least-squares problem⁶⁹,

$$\mathbf{G} \mathbf{m} = \mathbf{d}, \quad (2)$$

where $\mathbf{d}$ is the data vector containing observed vertical and horizontal deformation, $\mathbf{G}$ is the Green's function matrix describing the contribution of each source to each observation point, and $\mathbf{m}$ is a vector of source opening magnitudes.

To estimate the optimal source geometry, depth, and opening amplitude, we first determine source geometries by fitting to the InSAR-derived vertical deformation (Fig. S7). We then perform a grid search within the upper 1 km of the subsurface to identify the optimal depth of the opening source (Fig. S7). The amplitudes of opening are subsequently estimated through inversion of both vertical and horizontal deformation data (Fig. S7). Sources are assumed to be located directly beneath the deforming surface

and are approximated by polygonal geometries. Because Okada solutions are defined for rectangular sources, deformation due to unit opening of a 100 m by 100 m square at the centre depth of each layer is computed and assembled to represent each polygonal source. Model observation points are arranged on a regular grid with 100 m spacing, and InSAR-derived deformation fields are downsampled from 10 m to 100 m resolution to match this grid. Prior to inversion, the mean deformation outside all source geometries is subtracted from the observed vertical and horizontal deformation fields.

Elastic distributed-source model. The elastic distributed-source model is based on the poroelastic formulation of⁷⁴, which describes surface deformation induced by volumetric strain within a reservoir bounded by low-permeability layers. For an infinitely long source, surface vertical and horizontal displacements normal to the source are given by

$$u_{up} = \frac{2(1 + \nu_u)Bh\Delta\zeta}{3\pi} \left[\tan^{-1} \frac{x+a}{d} - \tan^{-1} \frac{x-a}{d} \right] \quad (3)$$

$$u_{hz} = \frac{(1 + \nu_u)Bh\Delta\zeta}{3\pi} \ln \frac{d^2 + (x+a)^2}{d^2 + (x-a)^2} \quad (4)$$

where ν_u is the undrained Poisson's ratio, B is Skempton's coefficient, h is the reservoir thickness, $\Delta\zeta$ is the change in volumetric fluid content, d is the reservoir depth, and a is the half-width of the reservoir.

All parameters except d and a are combined into a single amplitude coefficient $A = \frac{(1+\nu_u)Bh\Delta\zeta}{3\pi}$, which controls the magnitude of deformation. This formulation allows the poroelastic problem to be treated as an equivalent elastic distributed-source problem. We generalize this approach to three dimensions by allowing sources to take the form of polyhedrons⁷².

We assume a 100 m-thick layer to represent the aquifer within which distributed deformation takes place. As with the opening model, we first determine source geometries by fitting to the InSAR-derived vertical deformation and then perform a grid search within the upper 1 km of the subsurface to find the optimal depth (Figs. S7 and S5). Once this depth is found, we run an inversion to find the spatial distribution of deformation that fits both the vertical and horizontal deformation observations (Figs. S7 and S5). Surface deformation is attributed to volumetric strain within polygonal prism sources located in these layers beneath the deforming surface. Deformation due to unit volumetric strain of a cubic element is computed using Gauss-Legendre numerical quadrature, and individual cubes are assembled to represent each source geometry. Contributions from horizontal volumetric strain components (e_{11} and e_{22}) and vertical volumetric strain (e_{33}) are computed separately, with the vertical component weighted by a free parameter α . Inversion is performed using non-negative least squares for each layer and for a range of α values (Figs. S7 and S5).

Model evaluation

Model performance is evaluated using the mean and standard deviation of the misfit between modelled and observed deformation (Figs. S7 and S5). For each tested parameter combination, residuals are calculated for both vertical and horizontal deformation components, and their mean and standard deviation are used as quantitative measures of model fit. For elastic distributed-source models, this evaluation is repeated for a range of values of the strain partitioning parameter α .

Across all four study regions, models with source depths shallower than approximately 500 m consistently provide better fits to the observed deformation than deeper sources (Figs. S7 and S5). This preference for shallow sources is robust across both elastic opening and elastic distributed-source model configurations, indicating that the dominant contribution to the observed surface deformation arises from relatively shallow subsurface sources.

For the Uemachi Fault and Nara regions, residual statistics generally decrease with decreasing source depth and with increasing values of α in the

distributed-source models (Figs. S7 and S5). In these cases, models with larger α values yield residual means and standard deviations that are comparable to, or smaller than, those obtained from the best-fit elastic opening models. For the Oji and Arima–Takatsuki Fault Zone regions, models with $\alpha = 1$ also produce low residual means (Fig. S5), indicating that a range of strain partitioning scenarios can explain the observed deformation depending on local geometry.

We further assess the sensitivity of the elastic opening model to the assumed source geometry by examining its effect on the inferred source amplitude and depth. Although the estimated opening amplitudes and source depths vary with the assumed source geometry, the inversion results consistently indicate shallow source depths (e.g., < 300 m for Oji) (Fig. S10).

Data availability

Urban land uplift data for Greater Osaka are available at <https://doi.org/10.5281/zenodo.21270584>. Sentinel-1 SAR images are provided by the European Space Agency and accessed through the Alaska Satellite Facility <https://asf.alaska.edu/>. Groundwater level data for the Osaka Plain were obtained from the National Institute of Advanced Industrial Science and Technology https://gbank.gsj.jp/wellweb/GSJ_E/index.shtml and the Research Committee of Groundwater and Geo-environment, while data for the Kyoto Basin were obtained from the Ministry of Land, Infrastructure, Transport and Tourism <https://www.mlit.go.jp/en/>.

Code availability

All codes used to perform scientific analysis are open source. ISCE2 was used to generate stacks of coregistered SLC SAR images and is available at <https://github.com/isce-framework/isce2>, and archived at <https://doi.org/10.5281/zenodo.8157051>. Mintpy was used to generate InSAR time-series displacement and velocity products and is available at <https://github.com/insarlab>, and archived at 10.5281/zenodo.17695983. Unicycle was used to perform distributed source deformation models and is available at <https://bitbucket.org/sbarbot/unicycle/src/master/>. Hector was used to calculate velocity from GNSS time series data and is available at <https://teromovigo.com/product/hector/>. The scientific colour map vik was used to prevent visual distortion of the data, and is available at 10.5281/zenodo.8409685.

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Author contributions

S.F. and J.K. contributed equally to this study. J.K. and Y.K. designed the study; S.F. and J.K. generated the InSAR data; S.F., J.K. and Y.K. analysed the groundwater data; S.F. and S.B. conducted the modelling; S.F., J.K., Y.K. and S.B. analysed and discussed the results and wrote the manuscript.

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Competing interests

Sylvain Barbot is an Editorial Board Member for *Communications Earth and Environment* but was not involved in the editorial review or peer review, nor in the decision to publish this article.

Additional information

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